

Question 1

1 mark

101

Convert $-\frac{13\pi}{5}$ to degrees.

Question 2

a) 1 mark b) 1 mark c) 1 mark

102
103
104

a) From a group of 9 people, in how many ways can you select a committee of 4 members?

b) From a group of 9 people, in how many ways can you select a president, a vice president, a secretary, and a treasurer?

c) Explain why the answers in a) and b) are different.

Question 3

a) 2 marks b) 2 marks

105
106

A population of 500 bacteria will triple in 20 hours.

Using the formula given below,

$$A = Pe^{rt}$$

A = population after t hours

P = initial population

r = rate of growth

t = time in hours

a) Determine the rate of growth, r .

b) Determine how many hours it will take for the initial population to double with the same rate of growth.

Question 4

a) 1 mark b) 2 marks

107
108

Talla incorrectly solved the following trigonometric equation:

Solve: $2 \sec x - 5 = 0$; $0^\circ \leq x \leq 360^\circ$.

Talla's work:

$$2 \sec x - 5 = 0$$

$$\sec x = \frac{5}{2}$$

No solution, $\sec x$ cannot be greater than 1.

a) Explain her error.

b) Determine the correct solution.

Question 5

3 marks

109

Simplify the 6th term in the expansion of:

$$\left(2x - \frac{3}{x^2}\right)^{10}$$

Question 6

1 mark

110

Determine the arc length subtended by a central angle if the diameter is 19 cm and the central angle is 1.6 radians.

Note: A calculator is not required for the remaining test questions.

Question 7

4 marks

111

Solve the following equation algebraically for x , where $0 \leq x \leq 2\pi$.

$$2\cos^2 x = -3\sin x$$

Question 8

1 mark

112

In how many different ways can you arrange the letters in the word VOLLEYBALL?

State your answer as a factorial.

Question 9

2 marks

113

Is $(x - 2)$ a factor of the polynomial $p(x) = -x^4 - 3x^3 + 11x^2 + 3x - 10$?

Justify your response.

Question 10

1 mark

114

Determine the period of the sinusoidal function $y = \frac{1}{2} \sin\left(\frac{1}{3}x\right)$.

State your answer in radians.

Question 11

2 marks

115

The domain of $f(x)$ is $x \leq 2$. The domain of $g(x)$ is $x \geq -7$.

State the domain of $f(x) + g(x)$.

Justify your answer.

Question 12

3 marks

116

Prove the identity below for all permissible values of θ .

$$\frac{1}{1 + \cos \theta} = \csc^2 \theta - \frac{\cot \theta}{\sin \theta}$$

Left-Hand Side	Right-Hand Side

Question 13

1 mark

117

Explain how the end behaviours of the graphs of polynomial functions with an even degree and with an odd degree are different.

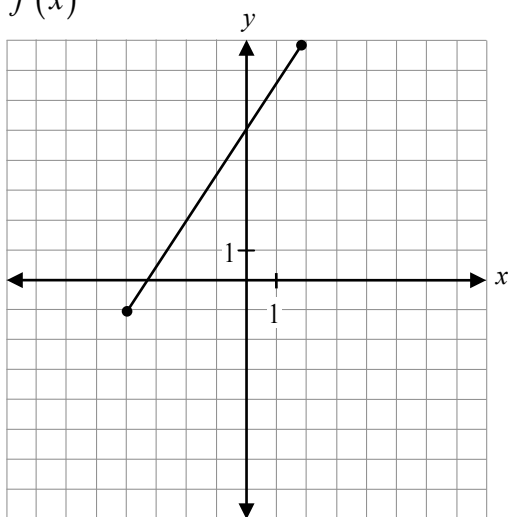
Question 14

2 marks

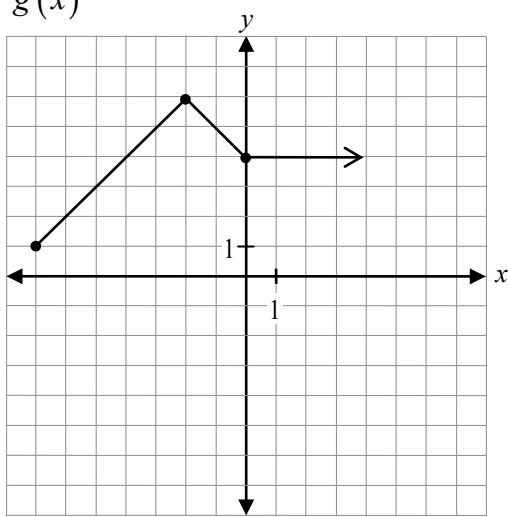
118

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $g(x) - f(x)$.

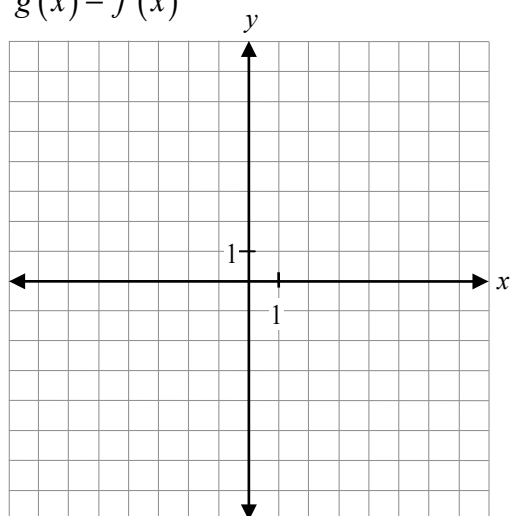
$f(x)$



$g(x)$



$g(x) - f(x)$



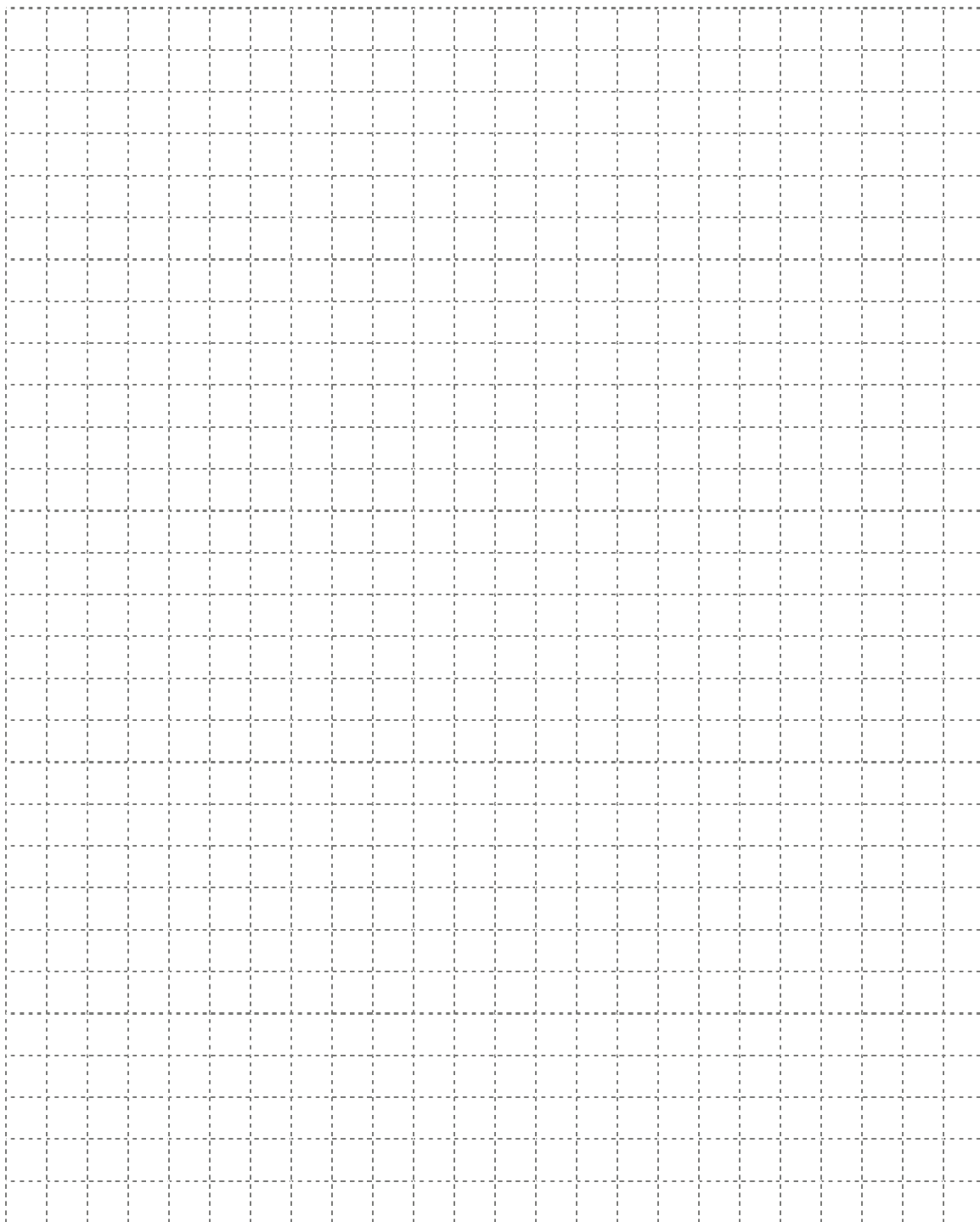
Question 15

2 marks

119

Given $f(x) = -3x + 7$, evaluate $f^{-1}(-2)$.

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